Module Code & Module Title

Level 5 – Network Operating Systems

Assessment Type

Logbook 9

Semester

2025/26 Spring/Autumn

Student Name: Bhumika Rayamajhi

London Met ID: 24045841

College ID: NP05CP4A240088

Assignment Due Date: Tuesday, December 23, 2025

Assignment Submission Date: Tuesday, December 23, 2025

Submitted To: Er Santosh Parajuli

I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Table of Contents

Logbook 8 1
1.1 Aim 1
1.2 Objectives 1
1.3 Required tools and concepts 2
1.4 Steps required for lab 3
1.5 Conclusion 14

Table of Figures

Figure 1: Creating directory as shown in the figure 3
Figure 2: Changing directory with relative pathname 3
Figure 3: Creating file with cat utility 4
Figure 4: grep command for searching line containing 'll' 5
Figure 5: grep command for searching line which do not contain 'll' 5
Figure 6: grep command to list whole line containing 'll' 6
Figure 7: Listing file which contains 'll' 6
Figure 8: non-case sensitive searching with grep 6
Figure 9: listing number of lines containing 'll' 7
Figure 10: Searching line that starts with k using grep 7
Figure 11: Listing lines of file testa with line number 7
Figure 12: Using alias without arguments 8
Figure 13: Removing the alias 8
Figure 14: Opening .bashrc 9
Figure 15: Preserving alias for next session 9
Figure 16: The alias from .bashrc 10
Figure 17:nwho alias command 10
Figure 18: Number of users account 10
Figure 19: History command 11
Figure 20: Re-executing the last but one command 12
Figure 21: re-executing previous command 13
Figure 22: Re-executing the last command which started with 'l' 13

Logbook 8

1.1 Aim

The aim of this lab is to acquire the skills of using basic UNIX command-line tools to manipulate files, search text and customize shells using practical tasks that included directory navigation, pattern matching and command history operations.

1.2 Objectives

The logbook assignment is given to year 2 students for Network Operating System module. This is the week 9 logbook assignment where students are required to learn basic UNIX command to manipulate files, search text and customize shells. It was done at the workshop of the module.

The objectives of this logbook are listed below:

- To make and access directory structures using UNIX commands by relative pathnames.
- To make text files with cat utility and search with grep with some options.
- To acknowledge the case-sensitive and case-insensitive searching and pattern matching methods.
- To develop temporary and permanent command aliases to make the command-line more efficient.
- To invoke command history capability to view and re-execute past prompted commands.

1.3 Required tools and concepts

The tools and concepts required for are:

Tools

- Debian Linux system to perform the given lab command.
- To make complex directory ‘mkdir’ command is used.
- Cat utility to generate new files and writing inside them.
- grep utility to search pattern and to filter text in files.
- To give shorter name for the command which are frequently used alias command is used.
- To see and control what has already been done, history command is used.
- fc command to re-execute commands entered into the history through events or patterns.

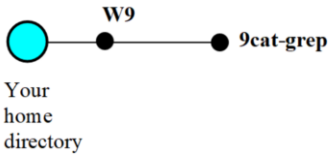
Concepts

- The concept of relative and absolute Pathnames to navigate the directory structures easily.
- Case Sensitivity concept to differentiate between lower case characters and uppercase characters while searching with grep.
- Shell Configuration Files in order to have command Aliases that can be persistent across sessions.
- Command History concept to re-execute and retrieve past commands effectively.\
- The concept of removing alias by unalias command.

1.4 Steps required for lab

First, open Debian terminal in your host OS to run command of Linux and follow these steps:

Creating the directory as shown in the figure



To create the directory shown in the figure, we have to write command "mkdir". "mkdir" is used to make directory. In order to achieve the directory shown in the picture we have to write "mkdir W9" and then "mkdir 9cat-grep".

```
bhumika@B:~$ mkdir W9
bhumika@B:~$ mkdir W9/9cat-grep
bhumika@B:~$ tree W9
W9
└── 9cat-grep

2 directories, 0 files
```

Figure 1: Creating directory as shown in the figure

Changing directory to 9cat-grep by one step using relative pathname

To change the directory using relative pathname we have to give relative pathname meaning from the parent directory of 9cat-grep not by the home. The command is "cd W9/9cat-grep".

```
bhumika@B:~$ cd W9/9cat-grep
bhumika@B:~/W9/9cat-grep$
```

Figure 2: Changing directory with relative pathname

Creating two files testa and testb using cat utility

testa	testb
Kkkll	KKKKK
lllmm	LLLLL
oo–oo	MMMMM
mmmdd	DDDDD
dddkk	

Creating two files using cat utility and writing the content shown in the picture above. The command to make file is "cat > file". These command lets you write inside the file. To escape we should click CTRL + D.

Figure 3: Creating file with cat utility

Writing the following grep commands

- grep ll testa
- grep -v ll testa
- grep -n ll testa
- grep -l ll *
- grep -i ll *
- grep -i LL *
- grep -c ll *
- grep '^K' testa testb
- grep -n '^' testa

The command "grep ll testa" return all the line with containing 'll'.

```
bhumika@B:~/W9/9cat-grep$ grep ll testa
Kkkll
lllmm
```

Figure 4: grep command for searching line containing 'll'

The command "grep -v ll testa" return all the line which do not contain 'll' in them.

```
bhumika@B:~/W9/9cat-grep$ grep -v ll testa
oo-oo
mmmdd
dddkk
```

Figure 5: grep command for searching line which do not contain 'll'

The command "grep -n ll testa" lists the whole line which contains 'll' as well as the line number.

```
bhumika@B:~/W9/9cat-grep$ grep -n ll testa
1:Kkkll
2:lllmm
```

Figure 6: grep command to list whole line containing 'll'

The command "grep -l ll *" lists file name which contains 'll' inside them.

```
bhumika@B:~/W9/9cat-grep$ grep -l ll *
testa
```

Figure 7: Listing file which contains 'll'

The command "grep -i ll *" list whole line with files which contains 'll' and acknowledge uppercase as well.

The command "grep -i LL *" list whole line with files which contains 'll' and acknowledge lowercase as well.

```
bhumika@B:~/W9/9cat-grep$ grep -i ll *
testa:Kkkll
testa:lllmm
testb:LLLLL
bhumika@B:~/W9/9cat-grep$ grep -i LL *
testa:Kkkll
testa:lllmm
testb:LLLLL
```

Figure 8: non-case sensitive searching with grep

The command "grep -c ll *" displays the number of lines containing 'll' in both testa and testb files.

```
bhumika@B:~/W9/9cat-grep$ grep -c ll *
testa:2
testb:0
```

Figure 9: listing number of lines containing 'll'

The command "grep '^K' testa testb" search for the line which begins with 'K'.

```
bhumika@B:~/W9/9cat-grep$ grep '^K' testa testb
testa:Kkkll
testb:KKKKK
```

Figure 10: Searching line that starts with k using grep

The command "grep -n '^' testa" shows every line with line number of testa file.

```
bhumika@B:~/W9/9cat-grep$ grep -n '^' testa
1:Kkkll
2:lllmm
3:oo-oo
4:mmmdd
5:dddkk
```

Figure 11: Listing lines of file testa with line number

Using alias without arguments

To make the alias as 'lsal', we have to give the command "alias lsal='ls -al' " and to show that the system store recently created alias we have to write the command 'alias' and it will show list of aliases. We can see there is an alias named 'lsal' which indicate that the system has stored the alias. After that using it in the home directory.

```
bhumika@B:~/W9/9cat-grep$ alias lsal='ls -al'
bhumika@B:~/W9/9cat-grep$ alias
alias ls='ls --color=auto'
alias lsal='ls -al'
bhumika@B:~/W9/9cat-grep$ lsal
total 16
drwxr-xr-x 2 bhumika bhumika 4096 Dec 22 15:00 .
drwxr-xr-x 3 bhumika bhumika 4096 Dec 22 14:45 ..
-rw-r--r-- 1 bhumika bhumika   30 Dec 22 15:00 testa
-rw-r--r-- 1 bhumika bhumika   24 Dec 22 15:00 testb
bhumika@B:~/W9/9cat-grep$ lsal /home/bhumika/W9
total 12
drwxr-xr-x 3 bhumika bhumika 4096 Dec 22 14:45 .
drwx------ 3 bhumika bhumika 4096 Dec 22 14:45 ..
drwxr-xr-x 2 bhumika bhumika 4096 Dec 22 15:00 9cat-grep
bhumika@B:~/W9/9cat-grep$
```

Figure 12: Using alias without arguments

Removing the alias

To remove the alias, we have to write the command "unalias lsal" and then to see 'lsal' alias don't exist anymore; we have to write alias to see the list of aliases.

```
bhumika@B:~/W9/9cat-grep$ unalias lsal
bhumika@B:~/W9/9cat-grep$ alias
alias ls='ls --color=auto'
bhumika@B:~/W9/9cat-grep$
```

Figure 13: Removing the alias

Defining the alias again and preserving it for next session

For defining the alias again and preserving it for next session, we have to write the command "nano .bashrc" to open .bashrc.

```
bhumika@B:~$ nano .bashrc
```

Figure 14: Opening .bashrc

After opening .bashrc, we have to add line "alias lsal='ls -al' " to preserve the alias for next session and then we have to click "CTRL+O and enter", and "CTRL+X" to save and exit from the .bashrc

```
bhumika@B: ~
  GNU nano 8.4                    .bashrc
    #alias fgrep='fgrep --color=auto'
    #alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=>

# some more ls aliases
#alias ll='ls -l'
#alias la='ls -A'
#alias l='ls -CF'

# Alias definitions.
# You may want to put all your additions into a separate file l>
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to en>
# this, if it's already enabled in /etc/bash.bashrc and /etc/pr>
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
alias lsal='ls -al'

                        [ Wrote 114 lines ]
^G Help     ^O Write Out^F Where Is ^K Cut      ^T Execute
^X Exit     ^R Read File^\ Replace  ^U Paste    ^J Justify
```

Figure 15: Preserving alias for next session

```
bhumika@B:~$ alias
alias ls='ls --color=auto'
alias lsal='ls -al'
bhumika@B:~$
```

Figure 16: The alias from .bashrc

Defining nwho alias for getting number of system file

To define nwho alias for getting the number of system file at UNIX computer, we have to write the command "alias nwho='getent passwd|wc -l' ".

```
bhumika@B:~$ alias nwho='getent passwd|wc -l'
```

Figure 17:nwho alias command

Displaying number of users account

To get the number of users account in the UNIX system, we have to write the command "nwho".

```
bhumika@B:~$ nwho
22
```

Figure 18: Number of users account

Displaying commands history

To get the history of commands, we can simply write history in the command. We can also limit the number of commands by giving number after writing history.

```
bhumika@B:~$ history 20
  402  grep -c ll *
  403  grep '^K' testa testb
  404  grep -n '^' testa
  405  ''alias lsal ='ls -al'
  406  alias lsal ='ls -al'
  407  alias lsal='ls -al'
  408  alias
  409  lsal
  410  lsal /home/bhumika/W9
  411  unalias lsal
  412  alias
  413  clear
  414  nano .bashrc
  415  tree
  416  clear
  417  nano .bashrc
  418  alias
  419  alias nwho='getent passwd|wc -l'
  420  nwho
  421  history 20
bhumika@B:~$
```

Figure 19: History command

Re-executing the last but one command

To re-execute last but one command without manually writing the command, we have to first give the command "fc -r". The last command in my case was history 20.

```
bhumika@B:~$ fc -r
history 20
  527  nano .bashrc
  528  clear
  529  W9
  530  cd w9
  531  history 20
  532  clear
  533  nano .bashrc
  534  alias
  535  alias nwho='getent passwd|wc -l'
  536  nwho
  537  history 2
  538  history 20
  539  clear
  540  history 2
  541  lsal
  542  ~/.bashrc
  543  tree
  544  nano .bashrc
  545  clear
  546  history 20
bhumika@B:~$
```

Figure 20: Re-executing the last but one command

Re-executing the command given three commands ago

To re-execute the command given three commands ago, we have to write exclamation mark subtracting three from it like "!-3".

```
bhumika@B:~$ !-3
nwho
22
```

Figure 21: re-executing previous command

Re-executing the previous command which begin with 'l'

To re-execute the previous command which begin with 'l', we have to write the command "fc -e- l".

```
bhumika@B:~$ fc -e- l
lsal
total 44
drwx------ 4 bhumika bhumika 4096 Dec 23 09:30 .
drwxr-xr-x 3 root    root    4096 Dec  4 07:15 ..
-rw------- 1 bhumika bhumika 5924 Dec 23 09:07 .bash_history
-rw-r--r-- 1 bhumika bhumika  220 Dec  4 07:15 .bash_logout
-rw-r--r-- 1 bhumika bhumika 3546 Dec 23 09:30 .bashrc
drwxr-xr-x 3 bhumika bhumika 4096 Dec 23 02:33 .local
-rw-r--r-- 1 bhumika bhumika  807 Dec  4 07:15 .profile
-rw-r--r-- 1 bhumika bhumika    0 Dec  4 07:15 .sudo_as_admin_su
ccessful
-rw-r--r-- 1 bhumika bhumika   31 Dec 23 09:23 testa
-rw-r--r-- 1 bhumika bhumika   25 Dec 23 09:24 testb
drwxr-xr-x 3 bhumika bhumika 4096 Dec 23 09:23 W9
bhumika@B:~$
```

Figure 22: Re-executing the last command which started with 'l'

1.5 Conclusion

This workshop enhanced my practicality in the use of UNIX utilities and shell customization to handle files and commands in a more effective manner. The practical activities were an in-depth exposure to the basics of command-line operations that underlie proper system management and the everyday administration of the UNIX work process.

I learned more about how small variations of the command can modify the output, how regular expression anchors can be used, and how temporary and permanent aliases can be defined in shell configuration files, by trying to make test files using cat utility via standard input redirection, experimenting with different grep options such as case-insensitive searching, line numbering, pattern inversion, and regular expression anchors, and understanding how the aliases and the history mechanism will enhance productivity. It also demonstrated the effectiveness of efficient command-line navigation and repeating operations without retyping: practicing command history features by the history command, re-executing past commands with event numbers by using fc and r commands, using negative integers as references to position and using patterns as a way of finding commands.

These skills will be valuable in future networking and server administration assignments, where text search speed, re-usable command shortcuts and good utilization of history is key to troubleshooting and automation. Fast log file filtering, command aliasing of complex operations, and command history capabilities will make life in system monitoring, configuration management, and regular administrative tasks much more productive in different IT infrastructure environments.